TEAM 1

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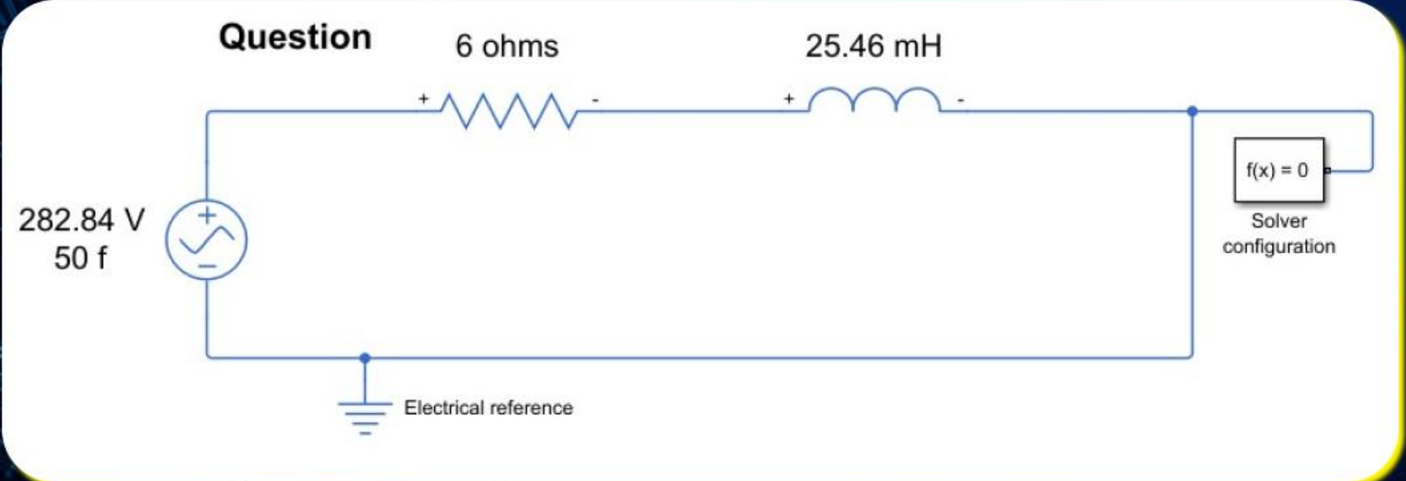
QUESTION:

1. A resistor of 6Ω and an ideal inductor 25.46mH are connected in series across 200V, 50 Hz supply. Determine
   i) Supply current
   ii) Active, Reactive and Apparent powers supplied by the source

   Obtain the waveforms of source voltage, supply current, voltage across the resistor and voltage across the inductor using SIMULINK and also validate the values of Active, Reactive and Apparent powers supplied by the source using SIMULINK.

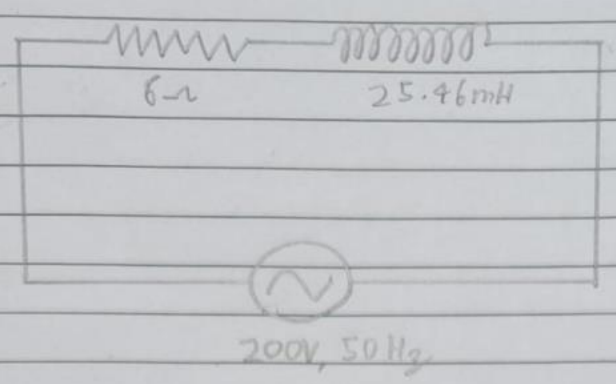


Given:

- Resistance, R = 6Ω
- Inductance, L = 25.46 mH
- Supply Voltage, V = 200V
- Frequency, f = 50 Hz

MANUAL SOLVING:

1.



Given:

$R = 6\,\Omega$

$L = 25.46\text{ mH}$

$V = 200\text{ V}$

$f = 50\text{ Hz}$

$$X_L = L\omega = L(2\pi f) = (25.46\times10^{-3})(2\pi\times50)$$

$$\approx 8\,\Omega$$

$$|Z| = \sqrt{R^2+X_L^2} = 10\,\Omega$$

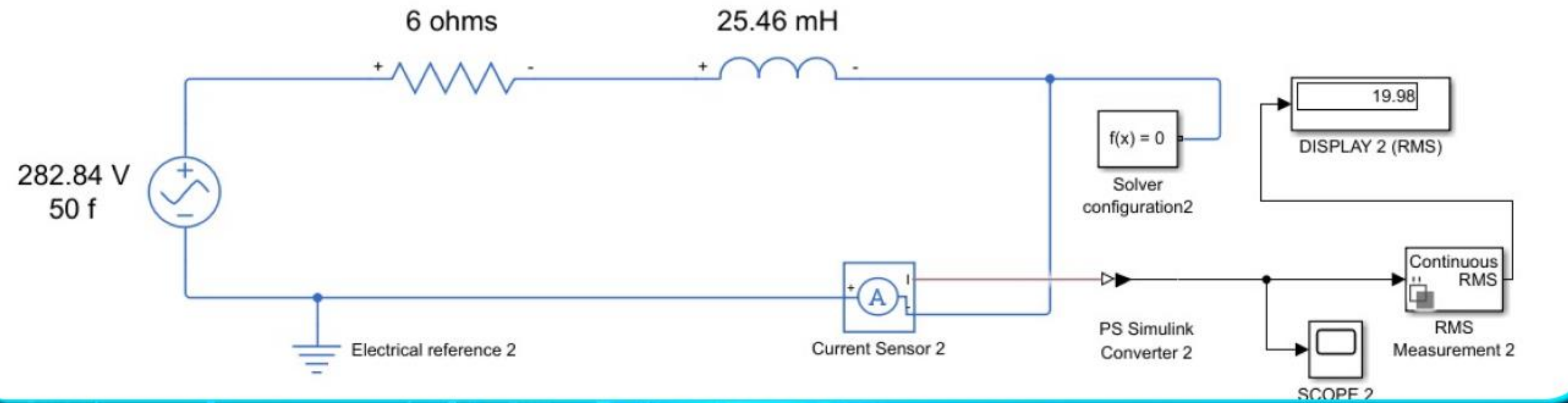
i) $$I = \frac{V}{|Z|} = \frac{200}{10} = 20\text{ A}$$

ii) $$P = I^2R = (20)^2(6) = \cancel{2} \; 2400\text{ W} = 2.4\text{ kW}$$

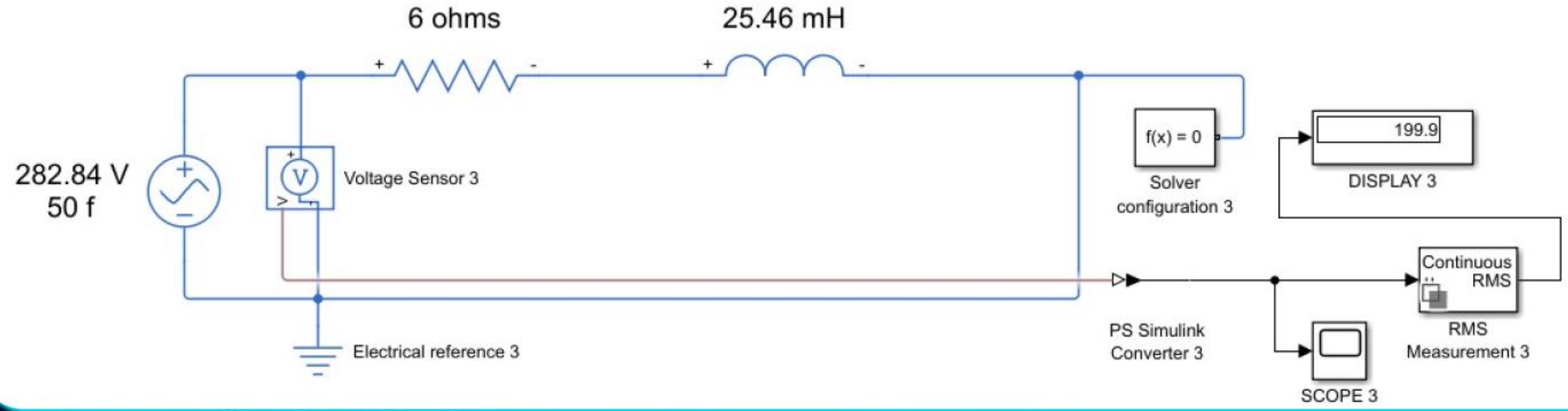
$$Q = I^2X_L = (20)^2(8) = 3200\text{ VAR} = 3.2\text{ kVAR}$$

$$S = \sqrt{P^2+Q^2} = \sqrt{(2.4)^2+(3.2)^2} = 4\text{ kVA}$$

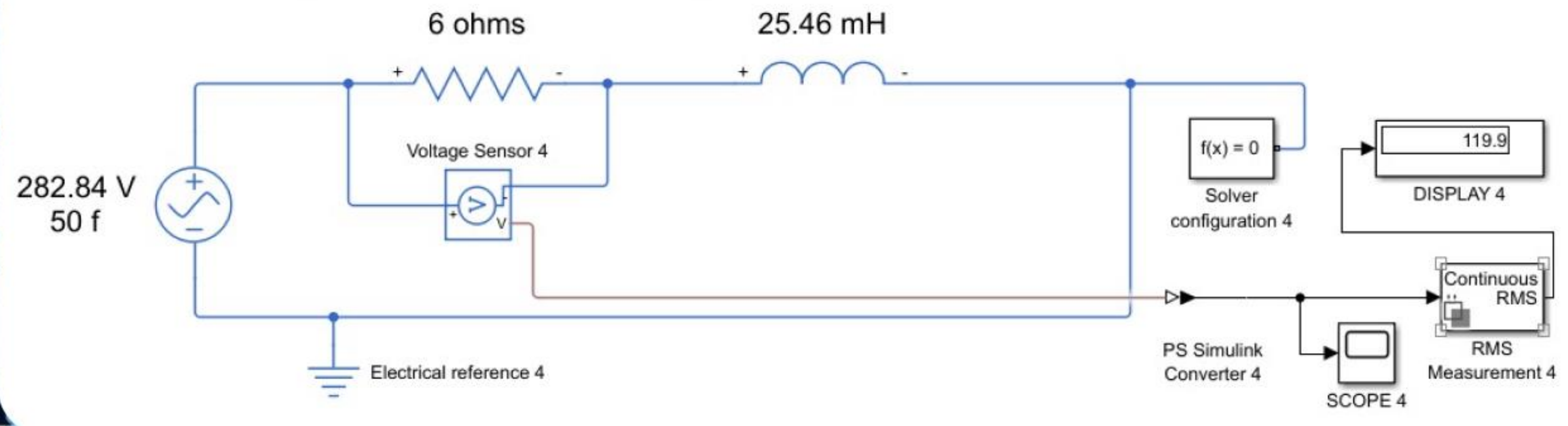
Supply Current (RMS)



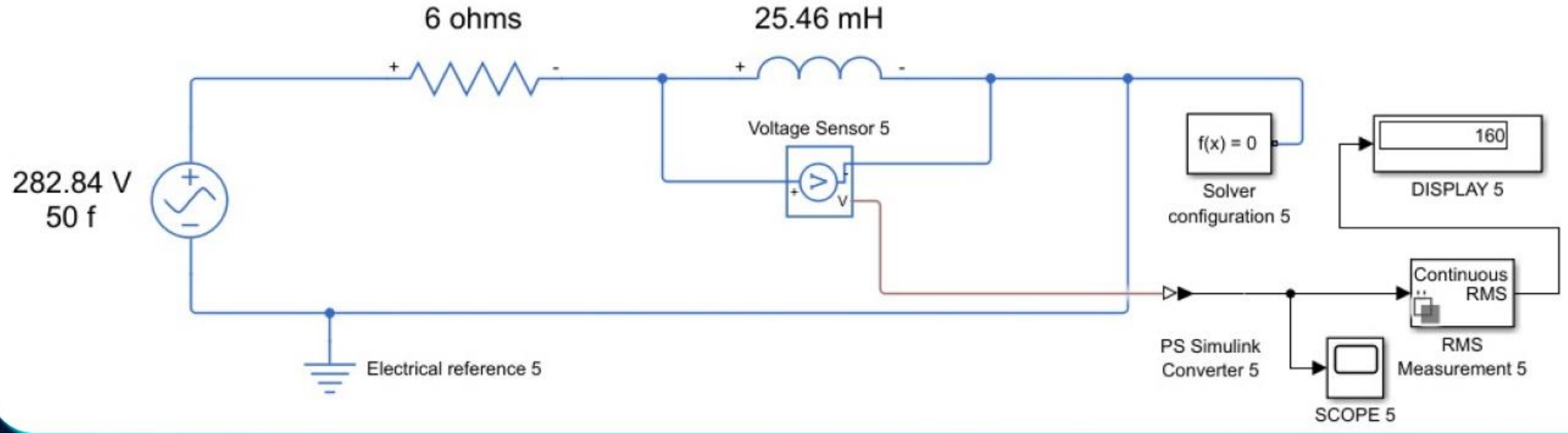
Source Voltage (RMS)



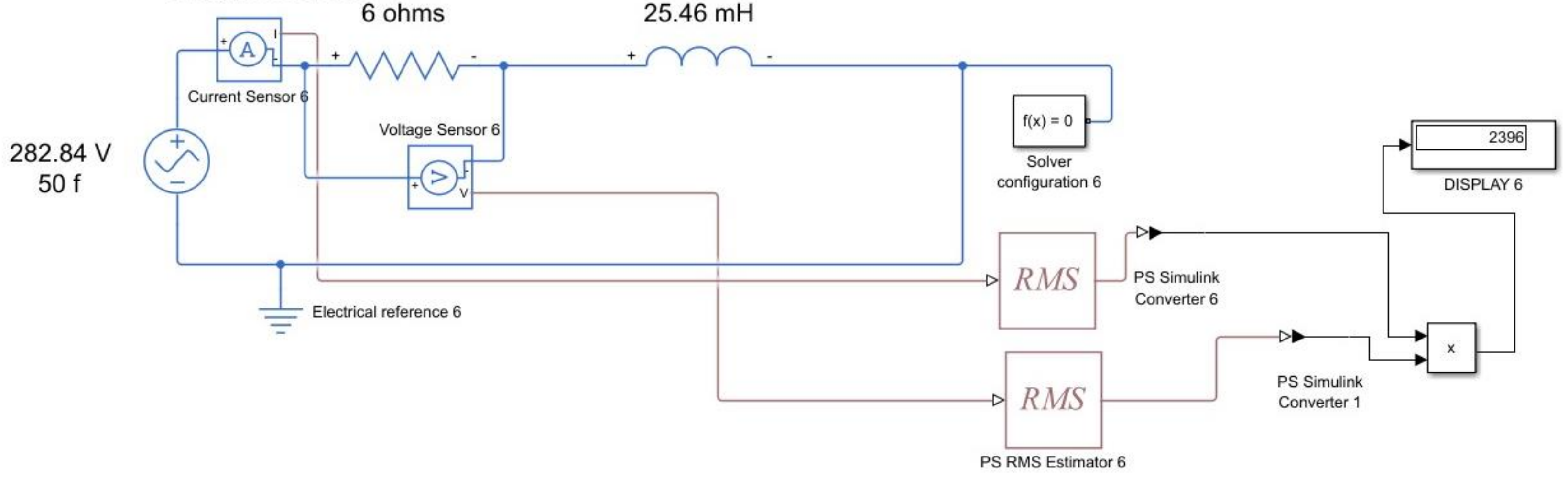
Voltage across resistor (RMS)



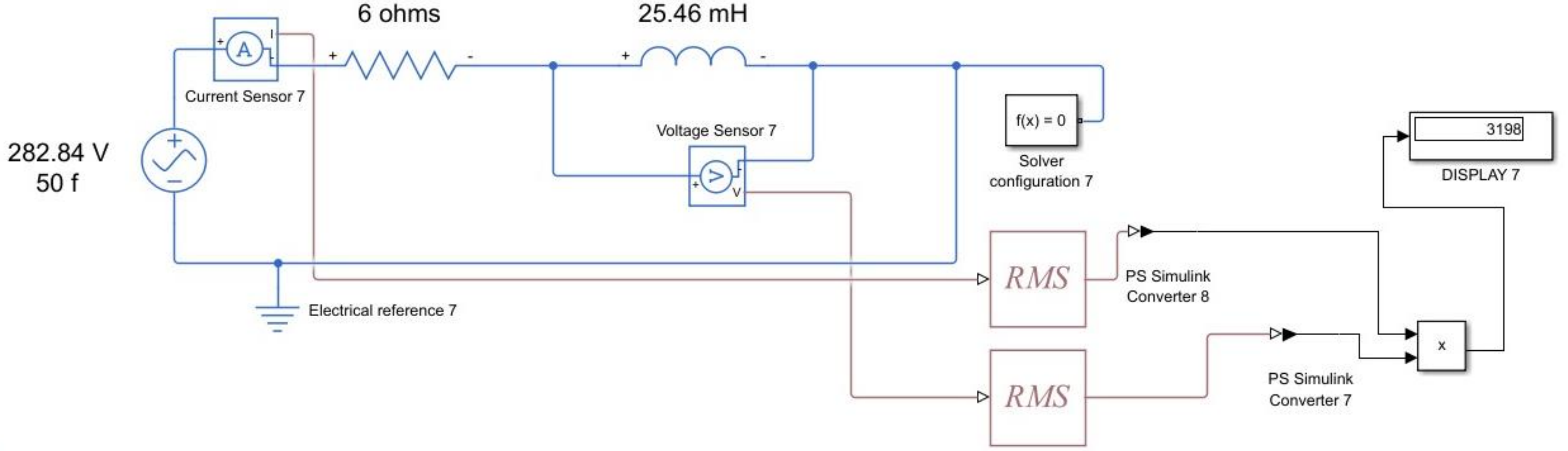
Voltage across inductor (RMS)



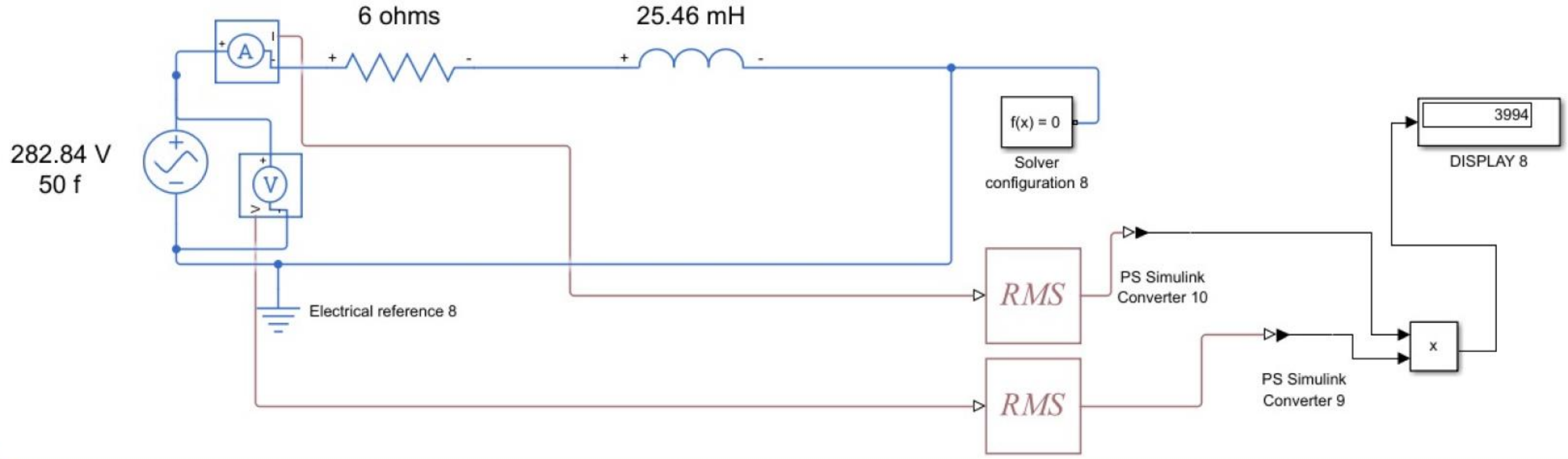
Active Power

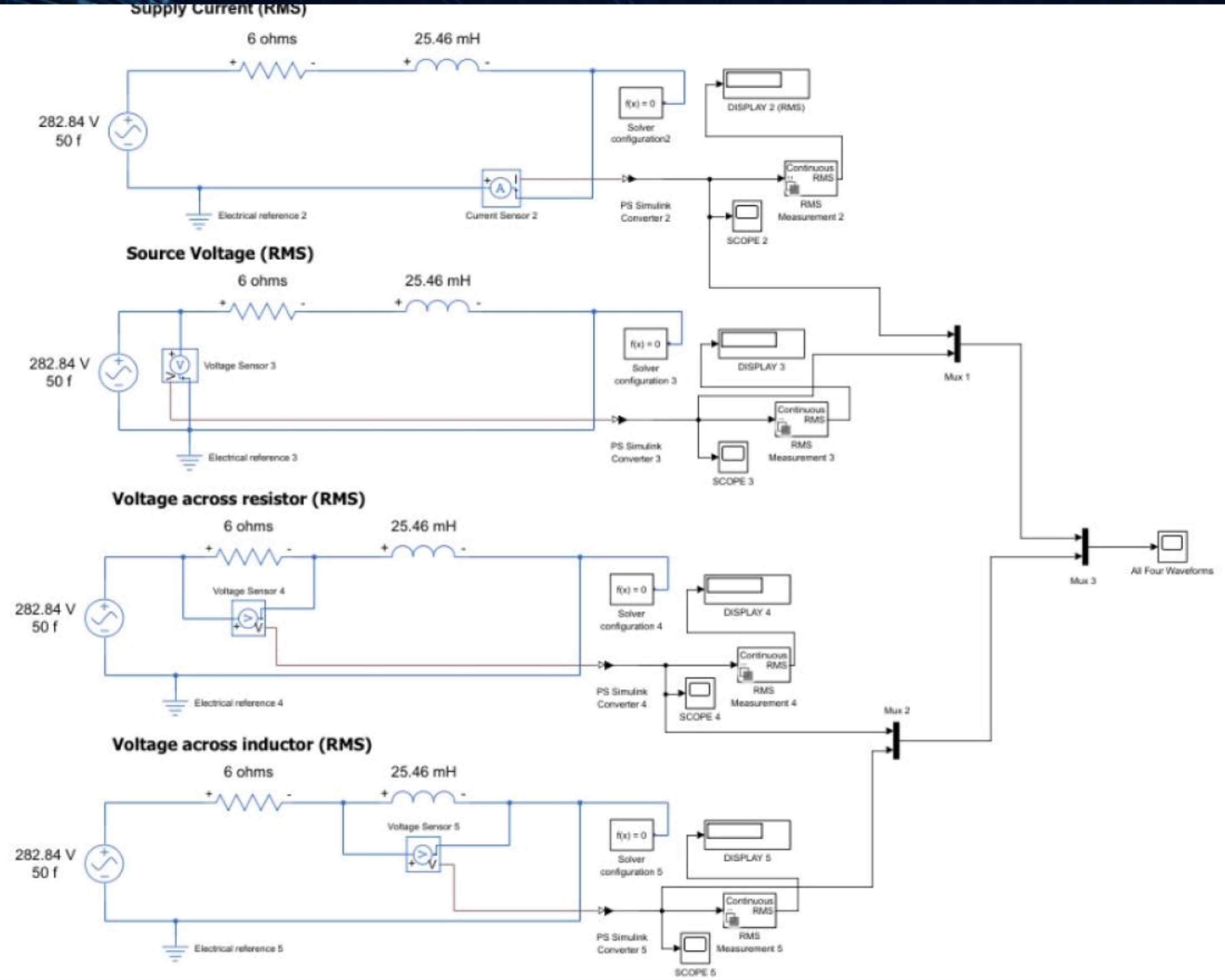


Reactive Power



Apparent Power




Supply Current (RMS)
6 ohms
25.46 mH
282.84 V
50 f
Electrical reference 2
Current Sensor 2
Solver configuration2
PS Simulink Converter 2
DISPLAY 2 (RMS)
Continuous RMS
RMS Measurement 2
SCOPE 2
Source Voltage (RMS)
6 ohms
25.46 mH
282.84 V
50 f
Voltage Sensor 3
Electrical reference 3
Solver configuration 3
PS Simulink Converter 3
DISPLAY 3
RMS Measurement 3
SCOPE 3
Mux 1
Voltage across resistor (RMS)
6 ohms
25.46 mH
282.84 V
50 f
Voltage Sensor 4
Electrical reference 4
Solver configuration 4
PS Simulink Converter 4
DISPLAY 4
RMS Measurement 4
SCOPE 4
Mux 2
Mux 3
All Four Waveforms
Voltage across inductor (RMS)
6 ohms
25.46 mH
282.84 V
50 f
Voltage Sensor 5
Electrical reference 5
Solver configuration 5
PS Simulink Converter 5
DISPLAY 5
RMS Measurement 5
SCOPE 5


350
300
250
200
150
100
50
0
-50
-100
-150
-200
-250
-300
-350
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1

Comparison of Manually Calculated Values vs Simulink Results

Parameter	Manually Calculated Value	Simulink Result (RMS)	Absolute Difference	Percentage Error (%)
Supply Current (I) (RMS)	20.00 A	19.98 A	0.02 A	0.10 %
Source Voltage (V_S) (RMS)	200.00 V	199.90 V	0.10 V	0.05 %
Voltage across Resistor (V_R) (RMS)	120.00 V	119.90 V	0.10 V	0.08 %
Voltage across Inductor (V_L) (RMS)	160.00 V	160.00 V	0.00 V	0.00 %
Active Power (P)	2.40 kW	2.396 kW	0.004 kW	0.17 %
Reactive Power (Q)	3.20 kVAR	3.198 kVAR	0.002 kVAR	0.06 %
Apparent Power (S)	4.00 kVA	3.994 kVA	0.006 kVA	0.15 %
Power Factor (cos φ)	0.60 (Lagging)	0.60 (Lagging)	–	–
Impedance (|Z|)	10.00 Ω	10.00 Ω	0.00 Ω	0.00 %
Inductive Reactance (X_L)	8.00 Ω	8.00 Ω	0.00 Ω	0.00 %